

AYUSH VELHAL

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EDUCATION

University of Texas at Dallas | Bachelor of Science in Computer Science, Minor in Business Administration

Expected May 2027

CERTIFICATIONS

AWS Certified Cloud Practitioner Demonstrated foundational knowledge of AWS cloud architecture, core services (EC2, S3, RDS, Lambda, VPC), cloud security principles, IAM access control, and monitoring using CloudWatch and CloudTrail.

CompTIA Security+ (SY0-701) In Progress — Preparing for the Security+ certification covering core cybersecurity principles including threat detection, network security, risk management, incident response, and security operations.

Full Stack Developer - Meta (Project Experience) Built a Django + MySQL full-stack application with REST APIs to automate high-volume authenticated requests, implement pagination and rate limiting, and store validated data in structured SQL formats for scalable processing.

Cybersecurity Professional Certified Certificate - Google Completed the 9-course cybersecurity certificate covering core security concepts and incident response.

Cybersecurity Analyst - Microsoft Completed the 9-course cybersecurity program covering security fundamentals, Azure security, identity management, network protection, and threat mitigation strategies.

Relevant Coursework

Relevant Coursework: Data Structures & Algorithms, Cybersecurity Risk Management, Operating Systems, Project Management, Professionalism and Communication in Business, Innovation and Entrepreneurship, Artificial Intelligence, Probability and Statistics for Computer Science, Database Systems, Computer Networks

SKILLS

Programming & Software Engineering: Java, Python, C++, Object-Oriented Programming (OOP), Data Structures & Algorithms, Debugging, Code Analysis

Backend & API Development: REST APIs, Backend Development, JSON Processing, Web Application Development

Databases & Data Management: SQL, PostgreSQL, SQL Queries, Relational Schemas, Database Design

Version Control & DevOps: Git, GitHub, CI/CD Concepts, Deployment Pipelines, Code Reviews

Cloud & Platforms: AWS (EC2, S3, RDS, Lambda), Google Cloud Platform (GCP - basic familiarity), Cloud Deployment

Systems & Scripting: Linux/Unix, Bash Scripting, Windows Scripting (basic), System Operations

Software Testing & Lifecycle: Unit Testing, Integration Testing, Software Development Lifecycle (SDLC), Agile (SAFe), Testing Frameworks

Tools & Collaboration: VS Code, Jupyter Notebook, Postman, Agile Workflows, Team Collaboration, Technical Documentation

LEADERSHIP & EXPERIENCE

Data Analyst | Global Career Accelerator

01/2026 — Present

- Built Python data pipelines (Pandas, Jupyter, Plotly) to ingest, clean, and transform 10K+ records per dataset for real-world analysis.
- Automated data processing and feature extraction, reducing manual data preparation time by ~40%.
- Analyzed multi-domain datasets (SQL + Pandas) using joins, filtering, and aggregations to uncover trends across 5+ datasets.
- Delivered dashboards and insights (Plotly), translating technical findings into actionable business recommendations.

Undergraduate Researcher | University of Texas at Dallas

10/2025 — 02/2026

- Conducted research on LLM-driven software testing, analyzing how Javadoc quality impacts test oracle generation across 10+ real-world Java codebases.
- Designed and ran controlled experiments using LLMs, evaluating effects of structured documentation on test accuracy and bug detection.
- Improved AI-generated test reliability by ~25–35% through enhanced documentation quality and structured comments.
- Built reproducible evaluation pipelines using Java, LLM APIs, and testing frameworks for automated test generation and validation.

Software Engineer | Kappa Theta Pi (KTP)

01/2025 — Present

- Developed full-stack features (React, Node.js) for app used by 50+ users, improving functionality and system usability.
- Optimized backend APIs + database queries, reducing response time by ~30% and improving data retrieval efficiency.
- Collaborated in Agile team (4–6 devs), accelerating feature delivery and reducing integration issues via code reviews.
- Built responsive UI components, improving user interaction speed and reducing friction across workflows.

Vice President of Technology | AWS Club

09/2025 — Present

- Led technical initiatives for 100+ members, organizing AWS/cloud workshops to improve hands-on learning and engagement.
- Delivered 5+ workshops on cloud architecture, APIs, and deployment, increasing technical skill adoption across members.
- Coordinated large-scale events, boosting participation and attendance by ~30%.
- Mentored students on projects + cloud tools, enabling development of real-world deployment-ready applications.

PROJECTS

Tark AI | Full Stack Hackathon Winning Project

04/2025

- Led development and deployment of an AI-powered document intelligence platform processing 100+ page financial reports, extracting key sections and reducing manual analysis time by ~80%.
- Built a scalable PDF parsing and NLP pipeline to structure unstructured data into searchable insights, improving accessibility and usability.
- Applied LLM-based summarization using OpenAI API to convert complex financial data into concise outputs for faster decision-making.
- Secured 2nd place at HackAI by delivering a fully deployed end-to-end AI system with an interactive interface under time constraints.

CareerOS AI | Full-Stack AI Platform

03/2026

- Built and deployed a full-stack platform supporting 100+ users, enabling real-time career chat + resume tracking with secure JWT authentication.
- Developed scalable backend (Node.js, Express, PostgreSQL), reducing API response latency to <3s under concurrent load.
- Integrated OpenAI GPT-4o, reducing manual resume review time by ~70% through instant AI feedback.
- Designed modern frontend (Next.js, Tailwind), enabling users to iterate resumes in <10 seconds, improving usability and engagement.

ResumeScanner | AI ML Project

03/2026

- Built a full-stack AI-powered ATS analyzer that processes resumes end-to-end and computes a detailed score, improving alignment by ~40%.
- Integrated OpenAI to generate context-aware bullet improvements and full resume rewrites, reducing editing time by ~70%.
- Designed a job matching system to detect missing keywords and compute match percentage with actionable insights in under 5 seconds.
- Developed and deployed a low latency cloud application with optimized PDF processing and secure APIs.

PosePerfectPro | Full-Stack Fitness Platform

03/2026

- Built and deployed a FastAPI and PostgreSQL backend to process workout videos and generate posture insights for 50+ sessions.
- Designed REST APIs for video upload, session tracking, and analytics with low latency performance.
- Integrated AI-based feedback and audio responses, reducing manual coaching effort by ~70%.
- Engineered a production-ready pipeline ensuring reliable performance under concurrent requests.